

Dear students

Here is an overview of the learning objectives for this module. By the end of the semester, you should be able to:

Introduction:

- Describe the color perception of the human eye
- Have a basic understanding of the Nyquist-Shannon sampling theorem
- Explain what digitization means
- List the advantages of digital images
- Give examples of direct and indirect imaging methods

Optical Imaging:

- Name the most important optical imaging methods used in preclinical biomedical research
- Identify, within a cell, specific structures and processes that can be imaged using molecular (optical) imaging techniques
- Explain how and for what purpose optical imaging is employed in drug development
- Compare the different optical imaging methods discussed regarding spatial resolution and sensitivity
- Explain how fluorescent signals are generated
- Explain why fluorescent probes used for in vivo imaging should have properties in the NIR range of the spectrum
- Give examples of fluorescent probes (low molecular weight probes and fluorescent proteins) that have their excitation/emission spectra in the NIR range and are typically used for in vivo imaging
- Explain the principles underlying confocal and multiphoton microscopy
- Provide examples of applications where intravital microscopy techniques are appropriate
- Explain how bioluminescence imaging works and for which types of experiments it is used

MRI:

- Understand where the MRI signal comes from
- Describe the main components of an MRI system
- Explain how spatial resolution is realized in MRI
- Explain the concept of Larmor frequency
- Identify safety considerations for operating an MRI
- Explain the concept of relaxation
- Describe the most important MR sequences

X-Ray, CT, PET:

- Explain how X-Rays are generated, including the two types of radiation
- Explain the term “tomography”
- Understand what a Sinogram is and how basic back-projection (“smear-back”) works
- Understand what Hounsfield Units are and explain the concept of “Windowing”
- Explain the basic principles of nuclear imaging (PET/SPECT)
- Explain the difference between PET and SPECT
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Optoacoustic Imaging:

- Explain why optoacoustic imaging is needed
- Describe how optoacoustic imaging is performed
- Explain how optoacoustics facilitates biological discovery
- Discuss whether optoacoustics can be used for medical diagnostics